

MNC Overall Report

User Guide

Configuring and running the workflow that consolidates weather, logistics, livestock, wildlife, and patrol data from EarthRanger into a single monthly report and dashboard for Mara North Conservancy

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Workflow id: **mara_north_event_report**

Overview

This guide walks you through configuring and running the MNC Overall Report workflow, which consolidates weather observations, logistics events, wildlife sightings, livestock events, and patrol data from EarthRanger into a single comprehensive monthly report and dashboard for Mara North Conservancy.

The workflow delivers, for each run:

- **7 weather charts** (HTML + PNG) — daily precipitation, temperature, wind speed, wind gusts, soil temperature, relative humidity, and atmospheric pressure per station
 - **Logistics tables** — balloon landing summary, airstrip operations (pivoted by arrival/departure), and airstrip maintenance log
 - **Livestock reports** — mobile boma movement map and summary, cattle count table, livestock predation map and summary, illegal grazing map
 - **Wildlife reports** — a wildlife incident map, and for all 8 species (elephant, buffalo, rhino, lion, leopard, cheetah, giraffe, hartebeest): a sightings map and summary table, with elephant and buffalo additionally getting a herd-composition map, herd-size bubble map, and herd-size bar chart
 - **Events overview** — total events recorded by date and by type, with a line chart
 - **Patrol reports** — foot, vehicle, motorbike, and overall patrol coverage maps, per-mode effort summaries, overall per-ranger effort, and conservancy patrol occupancy
 - **Results dashboard** — all 44 map, chart, and table widgets assembled into a single dashboard view
 - **1 Word report** (overall_report.docx) — generated from a Dropbox-hosted report template
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Prerequisites

Before running the workflow, ensure you have access to an **EarthRanger** instance with the following recorded for the analysis period:

- Weather station observations under the ER2ER - From GMMF subject group
- Logistics events: balloon_landing, airstrip_operations, airstrip_maintenance
- Livestock events: mobile_boma_rep, cattle_count, livestock_predation_rep, illegal_grazing_rep
- Wildlife incident events: snare_rep, fire_rep, wildlife_injury_rep, wildlife_treatment_rep, wildlife_carcass_rep
- Wildlife sighting events: elephant_sighting_rep, buffalo_sighting_rep, rhino_sighting_rep, lion_sighting_rep, leopardsightingrep, cheetah_sighting_rep, giraffe_sighting, hartebeest_sighting
- patrol_info events and associated patrol observations

You will also need network access to **Dropbox**, so the workflow can download the MNC conservancy boundary, parcels, and Word report template files at runtime.

Step-by-Step Configuration

Step 1 — Add the Workflow Template

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste the GitHub repository URL into the **GitHub Link** field, then click **Add Template**.

<https://github.com/wildlife-dynamics/mnc-overall-report.git>

Step 2 — Configure the EarthRanger Connection

Navigate to **Data Sources** and click **Connect**, then select **EarthRanger**. Fill in the connection form:

Field	Description
Data Source Name	A label to identify this connection (e.g. Mara North Conservancy)
EarthRanger URL	Your instance URL (e.g. your-site.pamdas.org)
EarthRanger Username	Your EarthRanger username
EarthRanger Password	Your EarthRanger password

Note: Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

Click **Connect** to save.

Step 3 — Select the Workflow

After the template is added, it appears in the **Workflow Templates** list as **mnc-overall-report**. Click the card to open the workflow configuration form.

Step 4 — Configure Workflow Details, Time Range, and EarthRanger Connection

The configuration form has three sections on a single page.

Set workflow details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes (e.g. reporting month or site)

Time range

Field	Description
Timezone	Select the local timezone (e.g. Africa/Nairobi UTC+03:00)
Since	Start date and time — all data from this point is fetched
Until	End date and time of the analysis window

Connect to ER

Select the EarthRanger data source configured in Step 2 from the **Data Source** dropdown (e.g. Mara North Conservancy).

Once all three sections are filled, click **Submit**.

Running the Workflow

Once submitted, the runner will:

- 1 Download the MNC community conservancy boundary and parcels files from Dropbox; repair invalid geometries; derive the conservancy AOI (`grazing_zone == Conservancy`), the non-conservancy grazing zones (coloured individually), and the Mara North Conservancy extent (used to compute the shared map zoom/centre); build reusable conservancy, grazing-zone, and parcels map layers.
- 2 Fetch weather station observations from the ER2ER - From GMMF subject group; extract seven meteorological variables from each observation; compute daily per-station aggregates; draw one line chart per variable.
- 3 Fetch all events once; split off balloon landing, airstrip operations, and airstrip maintenance events; clean and summarise each into a table.
- 4 Split off mobile boma, cattle count, livestock predation, and illegal grazing events; generate maps and summary tables — livestock maps are the only ones that additionally show the grazing-zone layer.
- 5 Split off wildlife incident events (snares, fires, carcasses, injuries, veterinary treatments); generate an incident map and summary tables.
- 6 Split off each of the 8 wildlife species' sighting events; generate a sightings map and summary table for each; elephant and buffalo additionally get herd-composition classification, a herd-size bar chart, and a herd-size bubble map.
- 7 Summarise all events by date and by type (excluding distance-count and airstrip-operations event types); draw a total-events line chart.
- 8 Fetch patrol_info events and associated patrol observations; convert to relocations; split into Foot, Vehicle, and Motorbike branches and convert each to trajectories using mode-appropriate segment filters; build a coverage grid and map for each mode and for the combined (overall) dataset; compute per-mode and per-ranger effort summaries and conservancy patrol occupancy.
- 9 Download the Word report template from Dropbox and assemble `overall_report.docx` from the outputs above.
- 10 Assemble all 44 widgets into the **MNC Overall Report dashboard**.
- 11 Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS`.

Note: Every task is automatically skipped if its input data is empty or an upstream step was skipped, so a run with no events of a given type simply omits that branch's outputs rather than failing. Widget-creation steps are the exception — they always run so that a placeholder widget still appears on the dashboard even if the branch behind it was skipped.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/`.

Weather

File	Description
<code>weather_summary_table.csv</code>	Daily per-station summary of all 7 weather variables
<code>precipitation_readings_over_time.html / .png</code>	Daily total precipitation

File	Description
temperature_readings_over_time.html / .png	Daily mean surface air temperature
wind_speed_readings_over_time.html / .png	Daily mean wind speed
wind_gusts_readings_over_time.html / .png	Daily maximum wind gusts
soil_temperature_readings_over_time.html / .png	Daily mean soil temperature
relative_humidity_readings_over_time.html / .png	Daily mean relative humidity
atmospheric_pressure_readings_over_time.html / .png	Daily mean atmospheric pressure

Logistics

File	Description
balloon_landing_summary_table.csv (+ HTML)	Passenger records by balloon company and lodge
airstrip_operations_summary_table.csv (+ HTML)	Client counts pivoted by camp/lodge and arrival/departure
airstrip_maintenance_summary_table.csv (+ HTML)	Dated log of airstrip maintenance activities

Livestock

File	Description
mobile_boma_movement_summary_table.csv	Daily count of boma movement events
boma_movement_map.html / .png	Boma movement locations map
total_cattle_count_summary_table.csv (+ HTML)	Cattle counts by zone, with total
total_livestock_predation_summary_table.csv	Daily count of livestock predation events
livestock_predation_summary_table.html	Rendered HTML of the daily predation count table (dashboard widget)
livestock_predation_summary_table.csv	Detailed per-event predation table (species, predator, total affected)
livestock_predation_events.html / .png	Predation events map, coloured by species
illegal_grazing_map.html / .png	Illegal grazing locations map

Wildlife Incidents

File	Description
wildlife_events_recorded.csv	Raw wildlife incident records (snares, fires, carcasses, injuries, veterinary treatments)
wildlife_incidents_summary_table.csv	Pivot table of incidents by type

File	Description
wildlife_incidents_recorded_by_date.csv	Daily count of unique incidents
wildlife_incidents_map.html / .png	Incident locations map, coloured by incident type

Wildlife Sightings

File	Description
elephant_sightings_events.html / .png	Elephant herd-composition sightings map
elephant_herd_types_map.html / .png	Elephant herd-size bubble map
elephant_herd_size_bar_chart.html / .png	Elephant herd-size distribution (5 bins)
overall_elephant_summary_table.csv (+ HTML)	Elephant sightings summary by herd type
buffalo_sightings_events.html / .png	Buffalo herd-composition sightings map
buffalo_herd_types_map.html / .png	Buffalo herd-size bubble map
buffalo_herd_size_bar_chart.html / .png	Buffalo herd-size distribution (5 bins)
overall_buffalo_summary_table.csv (+ HTML)	Buffalo sightings summary by herd type
lion_pride_sightings_map.html / .png	Lion sightings map, coloured by pride
overall_lion_summary_table.csv (+ HTML)	Lion sightings summary by pride
leopard_sightings_map.html / .png	Leopard sightings map, coloured by individuals present
overall_leopard_summary_table.csv (+ HTML)	Leopard sightings summary
cheetah_sightings_map.html / .png	Cheetah sightings map, coloured by individuals present
overall_cheetah_summary_table.csv (+ HTML)	Cheetah sightings summary
giraffe_sightings_map.html / .png	Giraffe sightings map
overall_giraffe_summary_table.csv (+ HTML)	Giraffe daily sightings summary
rhino_sightings_map.html / .png	Rhino sightings map
overall_rhino_summary_table.csv (+ HTML)	Rhino daily sightings summary
hartebeest_sightings_map.html / .png	Hartebeest sightings map
overall_hart_summary_table.csv (+ HTML)	Hartebeest daily sightings summary

Events Overview

File	Description
total_events_recorded_by_date.csv	All events recorded by date
total_events_recorded_by_type.csv	All events recorded by type
total_events_recorded.html / .png	All-events line chart

Patrols

File	Description
patrol_events.csv	Flattened patrol_info event details
patrol_purpose_summary.csv (+ HTML)	Patrol count by patrol purpose
patrol_relocations.geoparquet	Combined patrol observation dataset with patrol metadata
foot_patrol_efforts.csv	Foot patrol effort summary (count, distance, duration, speed)
foot_patrol_map.html / .png	Foot patrol coverage grid map
vehicle_patrol_efforts.csv	Vehicle patrol effort summary
vehicle_patrol_map.html / .png	Vehicle patrol coverage grid map
motorbike_patrol_efforts.csv	Motorbike patrol effort summary
motor_patrol_map.html / .png	Motorbike patrol coverage grid map
overall_patrol_map.html / .png	Combined (foot + vehicle + motorbike) patrol coverage grid map
patrol_trajectories.geoparquet	Reprojected overall patrol coverage grid
overall_patrol_efforts.csv (+ HTML)	Per-ranger summary of patrols, distance, and duration
patrol_coverage.csv (+ HTML)	Conservancy patrol occupancy percentage

Report

File	Description
overall_report.docx	Final Word report, assembled from the Dropbox-hosted template and all outputs above

Dashboard

The workflow run also produces the **MNC Overall Report dashboard**, viewable in the workflow runner, with 44 widgets:

Group	Widgets
Weather (7)	Precipitation, Temperature, Wind Speed, Wind Gusts, Soil Temperature, Relative Humidity, Atmospheric Pressure
Logistics (3)	Balloon Landing Summary, Airstrip Operations Summary, Airstrip Maintenance Summary
Livestock (5)	Mobile Boma Movement Map, Livestock Predation Events Map, Illegal Grazing Events Map, Total Cattle Count Summary, Livestock Predation Summary

Group	Widgets
Wildlife (21)	Wildlife Incident Map, Elephant Herd Size Map, Elephant Herd Composition Map, Elephant Herd Size Distribution, Elephant Herd Composition Summary, Buffalo Herd Size Map, Buffalo Herd Composition Map, Buffalo Herd Size Distribution, Buffalo Herd Composition Summary, Lion Sightings Map, Lion Sightings Summary, Leopard Sightings Map, Leopard Sightings Summary, Cheetah Sightings Map, Cheetah Sightings Summary, Giraffe Sightings Map, Giraffe Sightings Summary, Hartebeest Sightings Map, Hartebeest Sightings Summary, Rhino Sightings Map, Rhino Sightings Summary
Patrol (8)	Foot Patrol Coverage Map, Vehicle Patrol Coverage Map, Motorbike Patrol Coverage Map, Overall Patrol Coverage Map, Total Events Recorded, Patrol Purpose Summary, Overall Patrol Efforts, Conservancy Patrol Occupancy

Note: Each table widget is sortable and filterable in place; downloading directly from the widget is disabled — use the corresponding CSV output file for that.